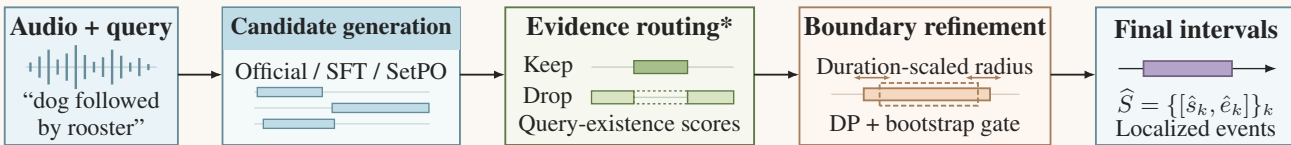


NOVA

Query-specific audio grounding

System overview



*Evidence routing is configuration-dependent.

Trains a candidate generator

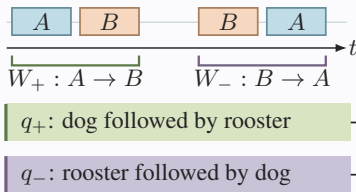
RelTwin

Locally valid timestamp contrasts

Training + diagnostic example

(a) Inverse-relation pairs

One audio x , shared A and B

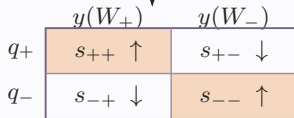


Both windows have acoustic support.
The query selects its window.

Inverse query $\implies$ exchanged labels

(b) Answer competition

Shared audio-language model
Frozen backbone + trainable LoRA



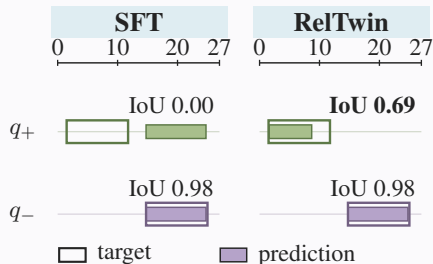
s : mean token log-likelihood

$$\mathcal{L}_{\text{cand}} = -\frac{1}{2}[\log p_+(+) + \log p_-(-)]$$

Query-conditioned answer preference

$$\mathcal{L} = \mathcal{L}_{\text{seq}} + \lambda \mathcal{L}_{\text{cand}} + \beta \mathcal{L}_{\text{replay}}$$

(c) Query-specific localization



Relation-development example:
shared time axis in seconds.